Lecture 12: Inverse matrix

A $n \times n$ matrix

Inverse of A: matrix B s.t.

$$AB = BA = I \leftarrow \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$

We write: $B = A^{-1}$.

Remarks:

① $Ax = b \iff x = A^{-1}b$

② A^{-1} doesn't always exist!
(For example, if $Ax = 0$ and $x \neq 0$)

③ A^{-1} exists $\iff$ A nonsingular
$\iff$ n pivots
$\iff$ independent cols
$\iff$ cols span $\mathbb{R}^n$
$\vdots$

④ $AB = I \qquad CA = I$

$$\Rightarrow \underbrace{(CA)}_{B}B = C\underbrace{(AB)}_{C}$$

⑤ $$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

⑥ $$\begin{bmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_n \end{bmatrix}^{-1} = \begin{bmatrix} 1/d_1 & & & \\ & 1/d_2 & & \\ & & \ddots & \\ & & & 1/d_n \end{bmatrix}$$

⑦ E_{ij} = subtracts ℓ times row j from row i

$$\underbrace{\begin{bmatrix} 1 & & \\ & 1 & \\ -\ell & & 1 \end{bmatrix}}_{E_{13}} \underbrace{\begin{bmatrix} 1 & & \\ & 1 & \\ \ell & & 1 \end{bmatrix}}_{E_{13}^{-1}} = \underbrace{\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}}_{I}$$

Prop: $(AB)^{-1} = B^{-1}A^{-1}$

Proof:
$$\begin{aligned}(AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} \\ &= AIA^{-1} \\ &= AA^{-1} \\ &= I\end{aligned}$$

$(B^{-1}A^{-1})(AB) = \cdots = I.$ ∎

A^{-1} by columns:

$$\underbrace{\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} & & \\ x_1 & x_2 & x_3 \\ & & \end{bmatrix}}_{A^{-1}} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{I}$$

$Ax_i = e_i$ (n systems)